

Red Hat Enterprise Linux Version 8



Important for Lab Exam of RHCE

- ❖ DHCP Server
- ❖ DNS Server
- ❖ PXE Server
- ❖ FTP Server
- ❖ Create User
- ❖ Samba Server
- ❖ 8e Linux LVM
- ❖ fd Linux raid auto
- ❖ 82 Linux swap / So

Command Used In Redhat Linux

#cat : this command is used to open file or directory
#vim : this command is used to open file or directory
#mkdir : this command is used to create directory file
#rm- rf : this command is used to remove forcefully
#mv : this command is used to move or rename file or directory
#chvt 1-7
#uname - show os
#uname -r : show version
4.18.0-193.el18.x86_64
4.18=kernel version
0-193=release

el8=distribution (enterprise linux)

x84_64=CPU architecture(x84=32bit)

pwd

Ctrl+Alt+F1=GUI

Ctrl+Alt+F2=GUI

Ctrl+Alt+F3-F6=CLI

Ctrl+Alt+F7=Show Services

tty

who

Installation:-

*** /boot**

/boot loader

-LILO

-GRUB

***/**

Root partition

*kernel (main kernel)

* GRUB (booting)

* initramfs(memory)

***/swap**

(Double of RAM)

Choose : Custom

Select : Standard Partition

luks=linux unified key system

"luks" is used to set password while opening system

vim /etc/sudoers

sudo fdisk -l

sudo /etc/shadow

#sudo cat /etc/shadow | grep santosh

#sudo adduser -d /santosh santosh

#hostname : this command is used to show hostname

#hostname -c : this command is used to show

passwd -d root : this is used to remove password from root user

passwd root : this command is used to set password in root user

Dig command:-

dig www.google.com

dig -x www.google.com

host www.google.com

Remove or Add DNS:-

#vim /etc/resolv.conf : this command is used to remove or add DNS

nameserver 8.8.8.8

nameserver 192.168.1.1

#route : this command is used to see kernel IP routing table / default gateway

Linux Distribution	
RPM:- • Redhat • Palo Alto • Cisco • Suse • Fedora • CentOS • Checkpoint	Debian:- • Kali • Ubuntu • Black Arc • Parrot

dhcp-server-4.3.6-40.el8.x86_64.rpm

- ❖ dhcp-server : name of package
- ❖ 4.3.6 : version
- ❖ 40 : release
- ❖ el8 :distribution enterprise linux
- ❖ x86_64 : architecture
- ❖ rpm : extension

#rpmquery dhcp: this command is used to check dhcp server installed or not
#rpmquery dhcp-server: this command show dhcp server details version
#rpm -q dhcp-server
#rpm -ql dhcp-server
#rpm -ql samba

#fdisk -l : this command is used to show partition disk summary
#parted /dev/nvme0n1 print free (parted is GNU command)
#fdisk /dev/nvme0n1

Welcome to fdisk (util-linux 2.32.1).

Changes will remain in memory only, until you decide to write them.

Be careful before using the write command.

Command (m for help): l

Command (m for help): q

Information of Partition stores in "fstab" (file system table):-

#vim /etc/fstab

#blkid :this command is used to show UUID

"blk" means block and Block devices are hardware devices

#mount -a : this command is used to check everything ok

File systems

- ❖ / : this is "root" and its file system is "XFS"
- ❖ /boot : this "boot" under "root" and its file system is "XFS"
- ❖ swap : this "swap" and its file system is "SWAP"

File sharing/server in linux through

- ❖ NFS
- ❖ FTP

File sharing in windows through

- ❖ SMB

Files sharing from windows to linux using SMB in linux known as SAMBA server

CIFS(common internet file system) knows both SMB and SAMBA

"Linux" is for Server Hosting

- ❖ FTP Server
- ❖ DNS Server
- ❖ SQL Server
- ❖ HTTP Server
- ❖ HTTPS Server
- ❖ PKI Server
- ❖ File Server
- ❖ Data Center Server

Host name in Linux is assigned as "Station123.example.com"

Host name is "Station123"

Domain name is "example.com"

Linux host name call or change from:-

- ❖ #etc/hosts
- ❖ network adapter

Show Hostname:-

#hostname : this command is used to show hostname

#hostnamectl : this command is used to show hostname

#cat /etc/hosts : this command is used to show all hostname

#cat /etc/hostname : this command is used to show hostname

Change Hostname:-

#vim /etc/hostname : this command is used to create hostname
station1.santosh.com

#hostname station1.santosh.com : this command is used to change hostname but
it will change temporally

hostnamectl set-hostname station1 : this command is used to change hostname

Verify hostname through using "ping" command

#ping station1.santosh.com

Change Hostname IPv4 or IPv6:-

#vim /etc/hosts : this command is used to change IP address

#systemctl restart NetworkManager : this command is used to restart network adapter

#systemctl restart systemd-hostnamed.service : this command is used to restart network adapter

"nmtui "

Network Manager text user interface

#nmtui : command is used for change hostname IP through GUI

#####

Network Adapter On or Off:-

#systemctl restart NetworkManager

Redhat main 90% file system file are located in "etc" folder

IP Address Gateway DNS Server IP Assigned

#vim /etc/sysconfig/network-scripts/ : this command is used for open

#cd /etc/sysconfig/network-scripts/

ll

#vim ifcfg-ens160

#ifup ens160 : Network Adapter On

#ifdown ens160 : Network Adapter Off

nmcli connection modify ens160 ipv4.addresses 192.168.1.111/24

You can change IP addresses, dns, gateway from above command

#nmcli connection show

#nmcli networking off

#nmcli networking on

#nmcli device status

```
#nmcli networking off; nmcli networking on
#bc : this is binary calculator
#init 0 : machine off
#init 6 : machine reboot
#nmcli connection show
# nmcli connection show
# nmcli device status
# nmcli connection down ens160
# nmcli connection modify ens160 connection.permissions user:garuna
# nmcli connection show ens160
# systemctl status NetworkManager
```

DNS

In starting time initial time DNS's all activity are done through host file. Host file is in both windows and linux DNS translate name to IP and IP to name in absence of DNS problem to run INTERNET

Query or Lookup:

- ❖ Recursive query/lookup : Local query eg:- ISP
- ❖ Iteration/Iterative query/lookup : WAN query eg:- root, dot/domain query
- ❖ Authoritative : If data are taken from original server
- ❖ Non-Authoritative : If data are not taken from original server

- Root DNS: world all "**name**" information here
- Master DNS
- Slave DNS
- Forwarding DNS/Caching-Only DNS

DNS Installing and Removing:-

```
#yum install bind-utils
```

```
#yum remove bind-utils : this is for removing
```

```
      : bind is DNS package
```

```
      : utils is utility
```

```
#yum install bind-chroot
```

```
#yum remove bind-chroot : this is for removing
```

(Note: removing command can remove "**dig**" command)

```
#rpmquery bind : this command show version of bind
```

(Now bind version is "9")

DNS main configuration file is in "etc"

```
#vim /etc/named.conf : this is used to go in its config files
```

```
options {
```

```
listen-on port 53 { 127.0.0.1; any ; };
```

```
listen-on-v6 port 53 { ::1; };
```

```
directory    "/var/named";
```

```
dump-file    "/var/named/data/cache_dump.db";
```

```
statistics-file "/var/named/data/named_stats.txt";
```



```
memstatistics-file "/var/named/data/named_mem_stats.txt";
secroots-file "/var/named/data/named.secroots";
recursing-file "/var/named/data/named.recursing";
allow-query {localhost; any; };
allow-query-cache {localhost; any; };
(Note: "any" should be made write)
```

Selinux:-

```
#ls -lZ /etc/named.conf
-rw-r-----. 1 root named system_u:object_r:named_conf_t:s0 1761 Jun 28 19:46 /etc/named.conf
#semanage fcontext -a -t named_conf_t /etc/named.conf
(Manage named_conf_t to selinux group)
```

Selinux **off** and **On**

```
# ls -lZ /etc/named.rfc1912.zones
-rw-r-----. 1 root named system_u:object_r:named_conf_t:s0 1029 Feb 16 04:36 /etc/named.rfc1912.zones
#semanage fcontext -a -t named_conf_t /etc/named.rfc1912.zones
```

Link creates in Selinux:

```
# ln -s /etc/named.conf /var/named/chroot/etc/named.conf
(ln -s = link -soft ,for create soft link)
```

(Soft(s) link will be made by name of "named.conf" in /chroot/etc)

named-checkconf /etc/named.conf : this command is used to see the soft link is created or not

systemctl restart named

Firewall allow:-

Port 53/udp

Port53/tcp

firewall-cmd --permanent --add-port=53/tcp

firewall-cmd --permanent --add-port=53/udp

firewall-cmd --reload

{The above process and commands are used to enable or set "caching-only DNS"}

"Kickstart"

anaconda-ks.cfg

This is the example of automation it allow automation It gives automatic installation but source file will not be received it will be install locally It provide automatic scripts

Load **ftp** and **vsftpd** server

```
# df -h
# cd /run/media/root/RHEL-8-6-0-BaseOS-x86_64/
# cd AppStream/
# ll
# cd Packages/
# ll
# rpm -ivh vsftpd-3.0.3-35.el8.x86_64.rpm
# rpm -ivh ftp-0.17-78.el8.x86_64.rpm
# rpmquery vsftpd
# rpmquery ftp
# rpm -qi vsftpd
# rpm -ql vsftpd
# cd /etc/vsftpd
-rw-----. 1 root root 5039 Jan  7 19:36 vsftpd.conf
# vim vsftpd.conf
    anonymous_enable=YES ( by default it is "NO" and make it "YES")
    listen=YES ( by default it is "No" and make it "YES")
    listen_ipv6=NO ( by default it is "YES" and make it "NO")
    (Note: both listen statuses must be vice versa)
# firewall-cmd --add-service=ftp --permanent
# firewall-cmd --reload
```

```
# cd /var/ftp/pub/
# mkdir -p KS
      p=permission apply like parent folder
# ll
      drwxr-xr-x. 2 root root 6 Jun 28 22:22 KS
# cp -r /root/anaconda-ks.cfg /var/ftp/pub/KS/
      (file copied from /root to /var/ftp/pub/KS)
# ll KS/
      -rw-----. 1 root root 1341 Jun 28 22:27 anaconda-ks.cfg
# cd KS/
# ll
# chmod 644 anaconda-ks.cfg
      -rw-r--r--. 1 root root 1341 Jun 28 22:27 anaconda-ks.cfg
# getsebool -a | grep ftp
# setsebool -P ftpd_full_access on
# systemctl restart vsftpd
# systemctl enable vsftpd
Created symlink /etc/systemd/system/multi-user.target.wants/vsftpd.service → /usr/lib/systemd/system/vsftpd.service.
# systemctl status vsftpd
      Active: active (running)
```

Create a New Virtual Redhat machine to applying above configuration

Press tab when Installing interface is shown and then writes some command after "quite" pressing spacebar:-

ks=ftp://192.168.1.100/pub/KS/anaconda-ks.cfg

Press "Enter" and new machine automatically completing its installation process

(Note: the above IP address is taken from configured machine)

DHCP Server

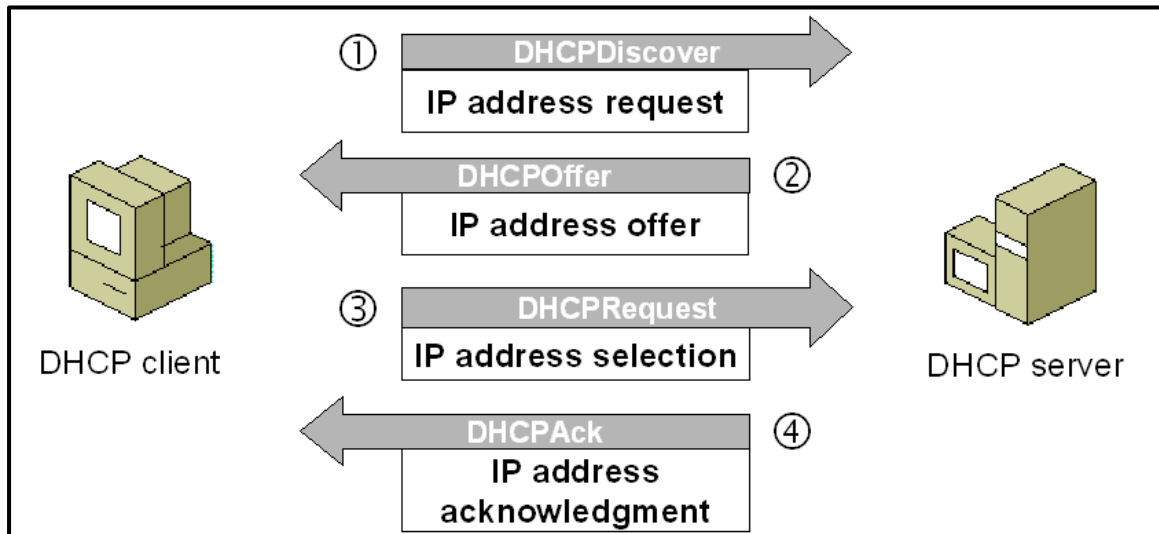
This is Open Protocol it is in linux, windows, cisco any many more. We can deploy dhcp server in router, switch, firewall, phone, modem, pc, NVR and etc... Generally dhcp server runs in IP network's UDP/67/68 protocols

And

IP/TFTP/UDP/69 this protocol runs in WDS, PXE Server and TFTP server

DHCP Client

This is Open Protocol it is in linux, windows, cisco any many more



DHCP was invented by IETF (Internet Engineering Task Force)

Its main work to reduce the effort of administrator and distribute IP address automatically and it is made by IETF

Server is 67 ports and Client is 68 ports

Server listening in 67 port number and Client listening in 68 port number

DORA Process:

- **Discover -Broadcast**
- **Offer -Unicast or Broadcast**
- **Request-Broadcast**
- **Acknowledge-Unicast or Broadcast**(primarily in unicast)

DHCP Configuration:

```
# df -h
```

```
# cd /run/media/root/RHEL-8-6-0-BaseOS-x86_64/
```

```
# cd BaseOs/  
# ll  
# cd Packages/  
# rpm -ivh dhcp-server-4.3.6-47.el8.x86_64.rpm
```

After loading dhcp its configuration file is in:

```
# vim /etc/dhcp/dhcpd.conf  
# cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf  
    (Copy dhcp server to /etc/dhcp/dhcpd.conf)  
# vim /etc/dhcp/dhcpd.conf
```

```
    # option definitions common to all supported networks...  
        option domain-name "example.com";  
        option domain-name-servers station1.example.com;  
        authoritative;  
        default-lease-time 600;  
        max-lease-time 7200;
```

(You have to change or keep **bold** things as per above)

This is a very basic subnet declaration.

```
    subnet 10.1.1.1 netmask 255.255.255.0 {  
        range dynamic-bootp 10.1.1.2 10.1.1.253;  
        option broadcast-address 10.1.1.255;  
        option routers 10.1.1.254;  
    }
```

(Change the IP manually as per the above range)

systemctl restart dhcpd

(Job for dhcpd.service failed because the control process exited with error code.

See "systemctl status dhcpd.service" and "journalctl -xe" for details.) = this type of messages appears if something error while something entry in "**dhcpd.conf**" **now the subnet is 10.1.1.0** (the above subnet is replaced by 10.1.1.0)

```
# systemctl restart dhcpd    (above)
```

(If everything is "OK" no any messages appears after hitting above command)

Allow firewall permanently in dhcp

```
# firewall-cmd --add-service=dhcp --permanent  
success
```

firewall-cmd --reload

success

systemctl enable dhcpd

Created symlink /etc/systemd/system/multi-user.target.wants/dhcpd.service → /usr/lib/systemd/system/dhcpd.service.

Linux File System:-

FAT NTFS OR HPFS- WINDOWS

EXT

EXT 2

EXT 3-JOURNALING FILE SYSTEMS

EXT 4

XFS -HP+ JOURNALING+64 bit-redhat-silicon graphic

JFS -IBM

BTRFS- Oracle

SWAP -

/----main folder top folder

/boot-----which has boot loader

/root-----root user home folder root

/bin---shared lib kernel

/dev

/bin

/home-----user folder home

/media-----

/opt---

/tmp

/etc

/mnt

/sbin----high level command

/usr----server file keep server data

#####

Disk partition disk slicing

fdisk -l

dh -h

dh -T

Linux Partitions:-

- Standard Partition-Fixed
- LVM Partition(Logical Volume Manager)-Extended

SATA

NVME

SSD Chip

Check free Space on disk:-

parted /dev/nvme0n1 print free

(nvme01 is disk name)

fdisk /dev/nvme0n2

Then press "p" for print: - this shows number of partition

Then press "n" for check partition: - this show partition types

P primary

e extended

Select (default is p):-p

Partition number (1-4, default 1):- 1

Select sector:

Last sector: +10GB

Created a new partition 1 of type 'Linux' and of size of 9.3 GiB

Check partition enter: p

(Partition appears)

Save above partition press "w"

cat /proc/partitions :-this command is used to see the information of kernel

partprobe :- this command is used to inform kernel (by default kernel don't know about created partition)

After creating partition we need to format it

mkfs -t ext4 /dev/nvme0n2p1

mkfs.ext4 /dev/nvme0n2p1

(We can use anyone from above command)

After formatting partition we need to do mount it

mkdir /disk1 :- make folder before mount

mount /dev/nvme0n2p1 /disk1 :- mount partition in created folder disk1

df -h :- Check disk information

df -T :- Check disk information with created file system name (ext4)

vim /etc/fstab

echo /dev/nvme0n2p1 /disk1 ext4 defaults 0 0>/etc/fstab

(do not use this command it will remove all created partition)

Recover disks

vim /etc/fstab :- this command is use to see or add below information

/dev/nvme0n1p1	/boot	xfs	defaults	0	0
/dev/nvme0n1p2	/	xfs	defaults	0	0
/dev/nvme0n2p1	/disk1	ext4	defaults	0	0

(Save above information and reboot machine)

File System

File system is a mechanism file folder save store access or Interaction between hard disk files and folder

- **Windows File System**

- ❖ FAT
- ❖ NTFS

- **Linux File System**

- ❖ EXT
- ❖ EXT2
- ❖ EXT3
- ❖ XFS
- ❖ BTFS
- ❖ EXT4
- ❖ SWAP

HDD:-

SAS:- More life then SATA and PATA it is used in Server

SATA HDD /SSD

sda
sdb
sdc and so on....

NVME (Next Generation SSD)

nvmen1
nvmen2
nvmen3 and so on.....

Partition

Basic disk

-MBR:-Master boot record

- 4 Partition

-3 primary

-1 extended

OR

-4 primary

Dynamic disk

-GPT:-

-132 partition

boot partition

/boot